

1 Basic Concepts

Experiment An **experiment** is a procedure that can be *infinitely repeated* and has a well-defined set of **outcomes**.

Outcome An **outcome** is a possible result of an experiment.

Random Experiment A **random experiment** is an experiment that has more than one possible outcome.

Sample Space The set of all possible outcomes of an experiment is called the **sample space** of the experiment.

Event An **event** is a subset of the sample space.

When the random experiment is performed and the outcome is in a certain event, we say that this event *has occurred*.

2 An Intuitive Definition of Probability

1. Repeat the experiment n times.
2. Count the number of times the event A occurs and denote it by $\mathcal{N}(A)$.

Then $\frac{\mathcal{N}(A)}{n}$ is called the *relative frequency* of A in the n repetitions, and the *probability* of A is defined as

$$P(A) := \lim_{n \rightarrow \infty} \frac{\mathcal{N}(A)}{n}$$

3 Set Theory

3.1 Some Notations

Events and *sets* are equivalent in this context. Thus usually we consider the *sample space* S to be the *universe* and have the following notations:

- The **null** or **empty** set: $\emptyset$
- A is a **subset** of B : $A \subseteq B$
- The **union** of A and B : $A \cup B$
- The **intersection** of A and B : $A \cap B$
- The **complement** of A in S : A'

Note that the *relative complement* of A in B can be obtained by

$$B \setminus A = B \cap A'$$

3.2 Some Concepts

3.2.1 Mutually Exclusive Events

Definition 3.1 (Mutually Exclusive Events).

$A_1, A_2, \dots, A_k$ are **mutually exclusive** $\iff$ They are **disjoint** sets $\iff A_i \cap A_j = \emptyset$ for all $i \neq j$.

3.2.2 Exhaustive Events

Definition 3.2 (Exhaustive Events).

$$A_1, A_2, \dots, A_k \text{ are } \mathbf{exhaustive} \iff A_1 \cup A_2 \cup \dots \cup A_k = S.$$

3.2.3 Countable Sets

Definition 3.3 (Countable).

A set is **countable** if there exists an *injection* from it into $\mathbb{N}$.

Remark. Note that a countable set can be either *finite* or *infinite*.

3.3 Some Laws

3.3.1 Distributive Laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Theorem 3.1 (Distributive Laws).

$$A \cup (B_1 \cup B_2 \cup \dots) = (A \cup B_1) \cup (A \cup B_2) \cup \dots$$

$$A \cap (B_1 \cap B_2 \cap \dots) = (A \cap B_1) \cap (A \cap B_2) \cap \dots$$

3.3.2 De Morgan's Laws

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

Theorem 3.2 (De Morgan's Laws).

$$(A_1 \cup A_2 \cup \dots)' = A_1' \cap A_2' \cap \dots$$

$$(A_1 \cap A_2 \cap \dots)' = A_1' \cup A_2' \cup \dots$$

3.3.3 Inclusion-Exclusion Principle

Theorem 3.3 (Inclusion-Exclusion Principle).

Let $A_1, A_2, \dots, A_n$ be finite sets and $|A_i|$ denotes the cardinality of the set A_i , then

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{\substack{J \subseteq \{1, 2, \dots, n\} \\ J \neq \emptyset}} (-1)^{|J|+1} \left| \bigcap_{j \in J} A_j \right|$$

4 Probability and Probability Function

4.1 Definition

Definition 4.1 (Probability).

Probability is a *real-valued set function* that assigns a number $P(A)$ to each event A in the sample space S . $P(A)$ is called the **probability** of A , and satisfies the following 3 properties (*Probability Axioms*):

- $P(A) \geq 0$
- $P(S) = 1$
- For a *countable* number of *mutually exclusive* events $A_1, A_2, A_3, \dots$,

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

4.2 Some Properties

4.2.1 Inclusion-Exclusion Principle

Theorem 4.1 (Inclusion-Exclusion Principle).

If $A_1, A_2, \dots, A_n$ are n events, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{\substack{J \subseteq \{1,2,\dots,n\} \\ J \neq \emptyset}} (-1)^{|J|+1} P\left(\bigcap_{j \in J} A_j\right)$$

4.2.2 A Frequently Used Trick

Theorem 4.2.

$$P(A) = P(A \cap B) + P(A \cap B')$$

4.3 Example

Let $S = \{e_1, e_2, \dots, e_m\}$ be a sample space, where each e_i is an outcome of the experiment. Let $N(A)$ denote the number of outcomes in the event A . It is easy to prove that the set function $P(A)$ defined as follows is a probability function:

- $P(\{e_i\}) = \frac{1}{m}$ for all i
- $P(A) = \frac{N(A)}{N(S)}$ for any event A

5 Probability Space

Definition 5.1 (Probability Space).

A **probability space** is a triple $(S, \mathcal{F}, P)$ that consists of:

- A **sample space** S , which is a set of all possible outcomes.
- An **event space** $\mathcal{F}$, which is a set of all possible events.
- A **probability function** $P : \mathcal{F} \rightarrow [0, 1]$ that satisfies the *Probability Axioms*.